Stochastic Processes

Homework 3: Complete Step-by-Step Solutions

Lecture 3 | 20 problems | 100 points | 2 optional bonus problems

Reading alignment

Sheldon M. Ross, *Introduction to Probability Models*, 10th ed.

- **Primary:** Chapter 4: introductory discrete-time Markov chains, states, transition matrices, and multistep probabilities.
- **Supporting:** Lecture 3: first-step analysis, probability trees, and conditional expectation of state-dependent rewards.

Instructor note. This guide follows the distributed Homework 3, Problems 1–20, in their original order and includes worked solutions to the two optional interview-style bonus questions. All 20 graded problems total 100 points. First-step equations explicitly count the first move; conditional probability and expectation are distinguished from independence. Numerical results are kept exact when convenient.

I. Probability Trees and Conditional Reasoning — 12 points

Problem 1. Independent two-round game (3 points)

(a) Recording the two rounds in order gives the four paths

$$\boxed{\{WW, WL, LW, LL\}}.$$

(b) Independence permits multiplication of the round probabilities:

$$\boxed{P(W_1 \cap W_2) = 0.6(0.6) = 0.36}, \qquad \boxed{P(L_1 \cap W_2) = 0.4(0.6) = 0.24}.$$

(c) A win in the second round occurs on either the WW or LW path. Since these paths are disjoint,

$$P(W_2) = P(W_1 \cap W_2) + P(L_1 \cap W_2) = 0.36 + 0.24 = \boxed{0.60}.$$

This agrees with the fixed 0.6 win probability for each independent round.

Problem 2. A dependent two-round game (3 points)

(a) The tree has first-step branches W and L, each with probability 0.5. From W, the second-step probabilities are $(0.8, 0.2)$ for (W, L); from L, they are $(0.3, 0.7)$. Multiplying along each branch yields

Path	WW	WL	LW	LL
Probability	$0.5(0.8)$	$0.5(0.2)$	$0.5(0.3)$	$0.5(0.7)$
	0.40	0.10	0.15	0.35

These four path probabilities add to 1.

(b) The law of total probability partitions on the first round:

$$\begin{aligned} P(W_2) &= P(W_2 \mid W_1)P(W_1) + P(W_2 \mid L_1)P(L_1) \\ &= 0.8(0.5) + 0.3(0.5) = 0.40 + 0.15 = \boxed{0.55}. \end{aligned}$$

(c) Bayes' rule divides the probability of WW by the probability of a second-round win:

$$P(W_1 \mid W_2) = \frac{P(W_1 \cap W_2)}{P(W_2)} = \frac{0.40}{0.55} = \boxed{\frac{8}{11} \approx 0.7273}.$$

Problem 3. Following paths for three rounds (3 points)

(a) The WLW path has a first win, then a loss following a win, then a win following a loss:

$$P(W_1 \cap L_2 \cap W_3) = 0.5(0.2)(0.3) = \boxed{0.03}.$$

(b) Similarly, the WWW path has probability

$$P(W_1 \cap W_2 \cap W_3) = 0.5(0.8)(0.8) = \boxed{0.32}.$$

(c) Given W_1, split on the two possible round-2 outcomes. The last result alone determines the next-step win probability:

$$\begin{aligned} P(W_3 \mid W_1) &= P(W_3 \mid W_2, W_1)P(W_2 \mid W_1) \\ &\quad + P(W_3 \mid L_2, W_1)P(L_2 \mid W_1) \\ &= 0.8(0.8) + 0.3(0.2) = 0.64 + 0.06 = \boxed{0.70}. \end{aligned}$$

The simplification $P(W_3 \mid W_2, W_1) = P(W_3 \mid W_2)$ uses the stated one-step dependence rule.

Problem 4. Conditional expected winnings (3 points)

(a) Condition on the current result:

$$\begin{aligned} P(W_{\text{next}}) &= P(W_{\text{next}} \mid W)P(W) + P(W_{\text{next}} \mid L)P(L) \\ &= 0.7(0.25) + 0.4(0.75) = 0.175 + 0.300 = \boxed{0.475}. \end{aligned}$$

Thus $P(L_{\text{next}}) = 1 - 0.475 = 0.525$.

(b) Let R be the next-round payoff. A win gives 10 and a loss gives -5, so

$$E[R] = 10(0.475) + (-5)(0.525) = 4.750 - 2.625 = \boxed{\$2.125}.$$

(c) **No.** The calculation uses conditional probabilities and total expectation. The win probability explicitly depends on the current result, so assuming independence would be inappropriate.

II. First-Step Analysis and Recursive Expectations 16 points

Problem 5. Waiting for a trading signal (4 points)

(a) From the inactive state, the first day always counts once. With probability 0.2 activation occurs immediately; with probability 0.8 the signal is still inactive and the expected additional wait is again m:

$$m = 0.2(1) + 0.8(1 + m) = \boxed{1 + 0.8m}.$$

(b) Subtract $0.8m$ from both sides:

$$0.2m = 1, \qquad \boxed{E[T] = m = 5 \text{ days}}.$$

(c) The event $T > 3$ means failing to activate on all three days. Each continuation from the inactive state has probability 0.8, hence

$$P(T \le 3) = 1 - P(T > 3) = 1 - (0.8)^3 = 1 - 0.512 = \boxed{0.488}.$$

Problem 6. Waiting for two consecutive heads (4 points)

(a) Define $m_0 = E[\text{additional flips until } HH \mid \text{state 0}]$ and $m_1 = E[\text{additional flips until } HH \mid \text{state 1}]$. After observing HH, the expected remaining time is 0.

(b) From state 0, a head advances to state 1 and a tail leaves us in state 0:

$$\boxed{m_0 = 1 + \tfrac{1}{2}m_1 + \tfrac{1}{2}m_0}.$$

From state 1, a head finishes the pattern, while a tail returns to state 0:

$$\boxed{m_1 = 1 + \tfrac{1}{2}(0) + \tfrac{1}{2}m_0 = 1 + \tfrac{1}{2}m_0}.$$

The leading 1 in each equation counts the very next flip, irrespective of its result.

(c) Rearrange the first equation: $\frac{1}{2}m_0 = 1 + \frac{1}{2}m_1$, or $m_0 = 2 + m_1$. Substitute the second:

$$m_0 = 2 + 1 + \tfrac{1}{2}m_0, \qquad \tfrac{1}{2}m_0 = 3, \qquad \boxed{m_0 = 6 \text{ flips}}.$$

It also follows that $m_1 = 1 + \frac{1}{2}(6) = 4$.

Problem 7. Waiting for the pattern HT (4 points)

(a) In state 0, H moves to state 1 and T leaves us in state 0. In state 1, T completes HT, while H leaves us in state 1: the latest head remains a valid start of the pattern.

(b) First-step conditioning gives

$$\boxed{m_0 = 1 + \tfrac{1}{2}m_1 + \tfrac{1}{2}m_0}, \qquad \boxed{m_1 = 1 + \tfrac{1}{2}m_1 + \tfrac{1}{2}(0)}.$$

(c) From the second equation, $\frac{1}{2}m_1 = 1$, so $m_1 = 2$. The first equation gives $m_0 = 2 + m_1$, and hence

$$\boxed{m_0 = 4 \text{ flips}}.$$

This is smaller than the six flips for HH in Problem 6. The two patterns are equally likely in a fixed two-flip window, but they overlap with earlier attempts in different ways.

Problem 8. Waiting for HTH (4 points)

(a) Transition descriptions are

State	next H	next T
0	1	0
1	1	2
2	complete	0

In particular, from state 1 another H leaves a trailing H, so the process stays in state 1.

(b) If the completed state has remaining time zero, conditioning on the next fair flip gives

$$m_0 = \boxed{1 + \tfrac{1}{2}m_1 + \tfrac{1}{2}m_0},$$
$$m_1 = \boxed{1 + \tfrac{1}{2}m_1 + \tfrac{1}{2}m_2},$$
$$m_2 = \boxed{1 + \tfrac{1}{2}(0) + \tfrac{1}{2}m_0}.$$

(c) The first equation says $m_0 = 2 + m_1$; the second says $m_1 = 2 + m_2$; and the third says $m_2 = 1 + \frac{1}{2}m_0$. Combining,

$$m_0 = 2 + 2 + 1 + \tfrac{1}{2}m_0 = 5 + \tfrac{1}{2}m_0.$$

Therefore $\frac{1}{2}m_0 = 5$, and

$$\boxed{m_0 = 10 \text{ flips}}.$$

The other values are $m_2 = 6$ and $m_1 = 8$.

III. Markov States and Transition Matrices 20 points

Problem 9. Profit/loss states (5 points)

(a) Let X_n denote the profit/loss result on day n, with state space $\boxed{S = \{W, L\}}$. The question describes a time-homogeneous one-step dependence rule.

(b) Order both rows and columns as (W, L). The first row lists next-day probabilities following W; the second row lists those following L:

$$\boxed{P = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix}}.$$

(c) $p_{LW} = 0.4$ is the probability tomorrow is profitable given today is unprofitable. $p_{WL} = 0.3$ is the probability tomorrow is unprofitable given today is profitable. Each row is a probability distribution:

$$0.7 + 0.3 = 1, \qquad 0.4 + 0.6 = 1.$$

All entries are nonnegative, so the matrix is valid.

Problem 10. A two-state weather chain (5 points)

(a) The possible states are sunny S and rainy R. For the order (S, R),

$$\boxed{P = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}}.$$

(b) A valid diagram has arrows $S \to S$ labeled 0.8, $S \to R$ labeled 0.2, $R \to S$ labeled 0.4, and $R \to R$ labeled 0.6. The first and last arrows are self-loops. The two outgoing probabilities at each state sum to one.

(c) Conditional on $X_0 = S$, the path $S \to R \to S$ requires rain tomorrow, then sun the next day. By the Markov property,

$$P(X_1 = R, X_2 = S \mid X_0 = S) = p_{SR}p_{RS} = 0.2(0.4) = \boxed{0.08}.$$

Problem 11. What information belongs in a state? (5 points)

(a) If the next flip depends on the last two outcomes, two histories ending in H (for example, HH and TH) can yield different probabilities for the next flip. Thus knowing only today's H does not generally give enough information for the one-step Markov property.

(b) Retain the ordered last two outcomes as the state:

$$\boxed{S' = \{HH, HT, TH, TT\}}.$$

This state description shifts the two-outcome window forward by one position after each new flip.

(c) From HH, an additional H produces HH and an additional T produces HT. From HT, the additional flip produces TH (if H) or TT (if T):

Current state	next H	next T
HH	HH	HT
HT	TH	TT

This is state augmentation: a sufficiently informative present state can replace the separate earlier history.

Problem 12. Recognizing the Markov assumption (5 points)

(a) **No.** For a Markov chain with present state only $X_2 = W$, the probability of $X_3 = W$ cannot change from 0.8 to 0.65 according to whether $X_1 = W$ or $X_1 = L$. The given conditional events have positive probability, so the discrepancy is genuine.

(b) The required equality would be

$$P(X_3 = W \mid X_2 = W, X_1 = W) = P(X_3 = W \mid X_2 = W, X_1 = L) = P(X_3 = W \mid X_2 = W).$$

But $0.8 \neq 0.65$, so it fails.

(c) A natural augmented state is the ordered pair

$$\boxed{Y_n = (X_{n-1}, X_n)}, \qquad Y_n \in \{WW, WL, LW, LL\}.$$

If the next-step law depends on no information beyond this pair, the augmented process has the Markov property. The data shown alone do not prove that no still earlier history matters.

IV. Multistep Transitions

24 points

Problem 13. Two steps in the win/loss chain (6 points)

(a) Starting from W, two paths end at W: $W \to W \to W$ and $W \to L \to W$. Their probabilities are $0.8(0.8)$ and $0.2(0.3)$, respectively:

$$P(X_2 = W \mid X_0 = W) = 0.64 + 0.06 = \boxed{0.70}.$$

(b) Multiply the matrix by itself, with each entry a sum over the two possible intermediate states:

$$\begin{aligned} P^2 &= \begin{pmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{pmatrix} \begin{pmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{pmatrix} \\ &= \begin{pmatrix} 0.8(0.8) + 0.2(0.3) & 0.8(0.2) + 0.2(0.7) \\ 0.3(0.8) + 0.7(0.3) & 0.3(0.2) + 0.7(0.7) \end{pmatrix} \\ &= \boxed{\begin{pmatrix} 0.70 & 0.30 \\ 0.45 & 0.55 \end{pmatrix}}. \end{aligned}$$

(c) The row-L, column-W entry is $p^{(2)}_{LW} = 0.45$: starting with an unprofitable day, the probability of a profitable state *after two transitions* is 45%.

Problem 14. Propagating a weather distribution (6 points)

(a) Start with the row distribution $\boldsymbol{\pi}_0 = (0.5, 0.5)$. Multiplying by P,

$$\boldsymbol{\pi}_1 = \boldsymbol{\pi}_0 P = (0.5(0.8) + 0.5(0.4),\ 0.5(0.2) + 0.5(0.6)) = \boxed{(0.6, 0.4)}.$$

Thus tomorrow has probability 0.6 of sunshine.

(b) Multiplication yields

$$P^2 = \begin{pmatrix} 0.8(0.8) + 0.2(0.4) & 0.8(0.2) + 0.2(0.6) \\ 0.4(0.8) + 0.6(0.4) & 0.4(0.2) + 0.6(0.6) \end{pmatrix} = \boxed{\begin{pmatrix} 0.72 & 0.28 \\ 0.56 & 0.44 \end{pmatrix}}.$$

(c) Either multiply $\boldsymbol{\pi}_1 P$ or $\boldsymbol{\pi}_0 P^2$:

$$\begin{aligned} \boldsymbol{\pi}_2 &= (0.6, 0.4)P \\ &= (0.6(0.8) + 0.4(0.4),\ 0.6(0.2) + 0.4(0.6)) \\ &= \boxed{(0.64, 0.36)}. \end{aligned}$$

The sunny probability changes from 0.60 after one step to 0.64 after two steps. This is a calculation about finite-step distributions, not a long-run classification claim.

Problem 15. Three-step transitions (6 points)

(a) From Problem 13, $P^2 = \left(\begin{smallmatrix} 0.70 & 0.30 \\ 0.45 & 0.55 \end{smallmatrix}\right)$. Multiply once more:

$$\begin{aligned} P^3 &= P^2 P \\ &= \begin{pmatrix} 0.70(0.8) + 0.30(0.3) & 0.70(0.2) + 0.30(0.7) \\ 0.45(0.8) + 0.55(0.3) & 0.45(0.2) + 0.55(0.7) \end{pmatrix} \\ &= \boxed{\begin{pmatrix} 0.650 & 0.350 \\ 0.525 & 0.475 \end{pmatrix}}. \end{aligned}$$

(b) With state order (W, L), the first column of P^3 gives the probabilities of reaching W:

$$\boxed{P(X_3 = W \mid X_0 = W) = 0.650}, \qquad \boxed{P(X_3 = W \mid X_0 = L) = 0.525}.$$

(c) For example, $p^{(3)}_{WW} = p^{(2)}_{WW} p_{WW} + p^{(2)}_{WL} p_{LW}$. The multiplication splits a three-step path after two steps, sums over the state then attained, and joins it to the last step. This is the Chapman–Kolmogorov principle.

Problem 16. Chapman–Kolmogorov with three states (6 points)

(a) Each entry of P is nonnegative, and the row sums are

$$0.5 + 0.5 + 0 = 1, \quad 0.25 + 0.5 + 0.25 = 1, \quad 0 + 0.4 + 0.6 = 1.$$

Therefore P is a valid transition matrix.

(b) Split on $X_1 \in \{A, B, C\}$. Starting at A, reaching C after two steps has conditional probability

$$\begin{aligned} P(X_2 = C \mid X_0 = A) &= p_{AA} p_{AC} + p_{AB} p_{BC} + p_{AC} p_{CC} \\ &= (0.5)(0) + (0.5)(0.25) + (0)(0.6) = \boxed{0.125}. \end{aligned}$$

The only positive-probability contributing path is $A \to B \to C$.

(c) The general Chapman–Kolmogorov formula is $p^{(m+n)}_{ij} = \sum_k p^{(m)}_{ik} p^{(n)}_{kj}$. In this case,

$$\boxed{p^{(2)}_{AC} = \sum_{k \in \{A,B,C\}} p_{Ak} p_{kC} = 0.125 = (P^2)_{AC}}.$$

V. Conditional Expectations and Synthesis 28 points

Problem 17. Expected trading payoffs (7 points)

(a) Starting at W, the next state is W with probability 0.8 and L with probability 0.2. Hence

$$E[r(X_1) \mid X_0 = W] = 10(0.8) - 5(0.2) = 8 - 1 = \boxed{\$7}.$$

(b) Starting at L, the next state is W with probability 0.3 and L with probability 0.7. Thus

$$E[r(X_1) \mid X_0 = L] = 10(0.3) - 5(0.7) = 3 - 3.5 = \boxed{-\$0.50}.$$

(c) The W row of P^2 is $(0.70, 0.30)$, so

$$E[r(X_2) \mid X_0 = W] = 10(0.70) - 5(0.30) = 7 - 1.5 = \boxed{\$5.50}.$$

(d) Let $R = r(X_1) + r(X_2)$. Linearity of conditional expectation gives

$$\begin{aligned} E[R \mid X_0 = W] &= E[r(X_1) \mid X_0 = W] + E[r(X_2) \mid X_0 = W] \\ &= 7 + 5.5 = \boxed{\$12.50}. \end{aligned}$$

No independence is needed for linearity; dependence is already represented by the transition matrix.

Problem 18. Weather-driven revenue and variance (7 points)

(a) With $X_0 = S$, the day-1 probabilities are $(0.8, 0.2)$. Thus

$$E[R_1 \mid X_0 = S] = 100(0.8) + 40(0.2) = 80 + 8 = \boxed{\$88}.$$

(b) Square the two possible revenue values and weight by their conditional probabilities:

$$E[R_1^2 \mid X_0 = S] = 100^2(0.8) + 40^2(0.2) = 8000 + 320 = \boxed{8320}.$$

Use the conditional variance identity:

$$\mathrm{Var}(R_1 \mid X_0 = S) = 8320 - (88)^2 = 8320 - 7744 = \boxed{576 \text{ dollars}^2}.$$

(c) The first row of P^2 from Problem 14 is $(0.72, 0.28)$. Therefore,

$$E[R_2 \mid X_0 = S] = 100(0.72) + 40(0.28) = 72 + 11.2 = \boxed{\$83.20}.$$

(d) Linearity of conditional expectation implies

$$E[R_1 + R_2 \mid X_0 = S] = 88 + 83.20 = \boxed{\$171.20}.$$

Revenue on consecutive days is generally dependent, but linearity still applies.

Problem 19. Reaching a target before bankruptcy (7 points)

(a) From capital 0, bankruptcy has already occurred; from 3, the target has already been reached. Thus

$$\boxed{h_0 = 0}, \qquad \boxed{h_3 = 1}.$$

(b) On the first round, capital $i \in \{1, 2\}$ moves to $i + 1$ or $i - 1$ with probability $1/2$. Therefore

$$h_i = \tfrac{1}{2} h_{i+1} + \tfrac{1}{2} h_{i-1}.$$

Using the boundary conditions,

$$\boxed{h_1 = \tfrac{1}{2} h_2}, \qquad \boxed{h_2 = \tfrac{1}{2} h_1 + \tfrac{1}{2}}.$$

(c) Substitute the first equation into the second:

$$h_2 = \tfrac{1}{2}(\tfrac{1}{2}h_2) + \tfrac{1}{2} = \tfrac{1}{4}h_2 + \tfrac{1}{2}.$$

Hence $\frac{3}{4}h_2 = \frac{1}{2}$, so $h_2 = \frac{2}{3}$ and

$$\boxed{h_1 = \tfrac{1}{2}(\tfrac{2}{3}) = \frac{1}{3}}.$$

Starting from \$1, the probability of reaching \$3 before \$0 is 1/3.

(d) Let $t_0 = t_3 = 0$ because no additional rounds are needed at a boundary. Each interior first-step equation counts one new round:

$$\boxed{t_1 = 1 + \tfrac{1}{2}t_2}, \qquad \boxed{t_2 = 1 + \tfrac{1}{2}t_1}.$$

Substitute the second into the first:

$$t_1 = 1 + \tfrac{1}{2}(1 + \tfrac{1}{2}t_1) = \tfrac{3}{2} + \tfrac{1}{4}t_1.$$

Thus $\frac{3}{4}t_1 = \frac{3}{2}$, giving $t_1 = 2$, and then $t_2 = 1 + \frac{1}{2}(2) = 2$. The expected duration from the stated starting capital is

$$\boxed{t_1 = 2 \text{ rounds}}.$$

Problem 20. A complete two-day trading forecast (7 points)

(a) Today's distribution is $(0.5, 0.5)$. Partition on today's state:

$$P(X_1 = W) = 0.5(0.7) + 0.5(0.4) = 0.35 + 0.20 = \boxed{0.55}.$$

Thus $P(X_1 = L) = 0.45$.

(b) Partition on day 1:

$$\begin{aligned} P(X_2 = W) &= P(X_2 = W \mid X_1 = W)P(X_1 = W) \\ &\quad + P(X_2 = W \mid X_1 = L)P(X_1 = L) \\ &= 0.7(0.55) + 0.4(0.45) = 0.385 + 0.180 = \boxed{0.565}. \end{aligned}$$

Thus $P(X_2 = L) = 1 - 0.565 = 0.435$.

(c) Let R_n be the day-n payoff. Weight the outcomes 100 and -50 by their day-specific probabilities:

$$\begin{aligned} E[R_1] &= 100(0.55) - 50(0.45) = 55 - 22.5 = \boxed{\$32.50}, \\ E[R_2] &= 100(0.565) - 50(0.435) = 56.5 - 21.75 = \boxed{\$34.75}. \end{aligned}$$

(d) By linearity,

$$E[R_1 + R_2] = E[R_1] + E[R_2] = 32.50 + 34.75 = \boxed{\$67.25}.$$

Daily independence is not assumed: the transition matrix handles the dependence, while expectation remains additive.

Bonus. Quant Interview Challenges optional, ungraded

Problem B1. Biased gambler's ruin (hedge-fund interview style)

Let h_i be the probability of reaching \$4 before \$0 from capital i. The boundary conditions are $h_0 = 0$ and $h_4 = 1$. First-step analysis gives, for $i = 1, 2, 3$,

$$h_i = 0.6h_{i+1} + 0.4h_{i-1}.$$

Writing the three interior equations and inserting the boundaries,

$$\begin{aligned} h_1 &= 0.6h_2, \\ h_2 &= 0.6h_3 + 0.4h_1, \\ h_3 &= 0.6 + 0.4h_2. \end{aligned}$$

Substitute the first and third equations into the second:

$$\begin{aligned} h_2 &= 0.6(0.6 + 0.4h_2) + 0.4(0.6h_2) \\ &= 0.36 + 0.24h_2 + 0.24h_2 = 0.36 + 0.48h_2. \end{aligned}$$

Therefore $0.52h_2 = 0.36$, so

$$\boxed{h_2 = \frac{0.36}{0.52} = \frac{9}{13} \approx 0.6923}.$$

Starting from \$2, the target is reached before bankruptcy with probability about 69.23%. The unequal up/down probabilities enter only through the first-step weights.

Problem B2. Three heads in a row (hedge-fund interview style)

Use the states: 0 = no trailing heads, 1 = one trailing head, 2 = two trailing heads, and completed = HHH observed. Let m_i be the expected additional flips until completion from state i; the completed state has remaining time 0. The next-flip transitions are

Current state	H	T
0	1	0
1	2	0
2	completed	0

The first-step equations are

$$\begin{aligned} m_0 &= 1 + \tfrac{1}{2}m_1 + \tfrac{1}{2}m_0, \\ m_1 &= 1 + \tfrac{1}{2}m_2 + \tfrac{1}{2}m_0, \\ m_2 &= 1 + \tfrac{1}{2}(0) + \tfrac{1}{2}m_0. \end{aligned}$$

The first equation gives $m_0 = 2 + m_1$, while the third gives $m_2 = 1 + \frac{1}{2}m_0$. Insert both into the second:

$$m_1 = 1 + \tfrac{1}{2}(1 + \tfrac{1}{2}m_0) + \tfrac{1}{2}m_0 = \tfrac{3}{2} + \tfrac{3}{4}m_0.$$

Consequently,

$$m_0 = 2 + \tfrac{3}{2} + \tfrac{3}{4}m_0, \qquad \tfrac{1}{4}m_0 = \tfrac{7}{2}.$$

Therefore,

$$\boxed{m_0 = 14 \text{ flips}}.$$

The intermediate expectations are $m_2 = 8$ and $m_1 = 12$. The argument requires only the same state design and first-step conditional expectations used for HH and HTH.